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Efficacy and Safety of Tranexamic Acid Combined with Absorbable Hemostat in Reducing Perioperative Blood Loss in Total Knee Arthroplasty

A Prospective, Blinded, Randomized Controlled Trial

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Background: Tranexamic acid (TXA) and absorbable hemostat (AH) are widely used to reduce perioperative blood loss in total knee arthroplasty (TKA). However, the efficacy of single-method hemostatic strategies is increasingly insufficient for meeting clinical demands. This study tested the efficacy and safety of TXA combined with AH for perioperative blood management in TKA.

Methods: Hemostatic efficacy was preliminarily evaluated through in vitro dynamic coagulation assays, lactate dehydrogenase activity measurements, and scanning electron microscopy, as well as in vivo using a rabbit liver bleeding model. The in vivo biocompatibility was also measured. Subsequently, the efficacy and safety of TXA combined with AH were further evaluated in a prospective, blinded study involving 149 individuals who were randomized to receive TXA, AH, or TXA+AH during TKA. The primary outcomes were perioperative blood loss, blood transfusion, hemoglobin and hematocrit levels, maximum hemoglobin change, anemia, and postoperative complications. Secondary outcomes included perioperative inflammation, coagulation function, and knee joint function.

Results: Coagulation assays and the liver hemostasis model demonstrated that TXA combined with AH effectively promoted coagulation, with satisfactory biocompatibility. The clinical results of 114 Han Chinese (East Asian) patients indicated that the combination significantly reduced perioperative blood loss in TKA (564.51 ± 136.26 mL in the TXA+AH group, 879.35 ± 85.62 mL in the TXA group, and 692.70 ± 96.06 mL in the AH group; $p < 0.001$) without an increase in thromboembolic events or wound-related complications. Additionally, the combination accelerated early postoperative knee function recovery without significantly affecting pain scores or inflammatory markers.

Conclusions: The combination of TXA and AH effectively reduced perioperative blood loss in TKA, accelerated early patient recovery, and did not increase the rate of complications.

Level of Evidence: Therapeutic Level I. See Instructions for Authors for a complete description of levels of evidence.

*Qing-Yi, MD, and Gang Wang, MM, contributed equally to this work.

Disclosure: This study was jointly supported by the National Key Research and Development Program of China (No. 2023YFB4606700), Clinical Research Incubation Program of West China Hospital (No. 2024HXFH008), Key Research and Development Program of the Sichuan Science and Technology Department (No. 23ZDYF2641), Cadre Health Research Program of Sichuan Province (No. 2023-118), and International Science and Technology Cooperation Program of Chengdu Science and Technology Bureau (No. 2023-GH02-00075-HZ). The **Disclosure of Potential Conflicts of Interest** forms are provided with the online version of the article (<http://links.lww.com/JBJS/I788>).

A **data-sharing statement** is provided with the online version of the article (<http://links.lww.com/JBJS/I790>).

Knee osteoarthritis is a chronic degenerative disease that severely impacts patient quality of life as the disease progresses to advanced stages^{1,2}. At present, total knee arthroplasty (TKA) is considered the gold-standard treatment for end-stage osteoarthritic conditions^{3,4}. However, it often necessitates extensive soft-tissue dissection and cancellous bone osteotomy, resulting in substantial perioperative blood loss of up to 800 to 1,000 mL, which is approximately 10% to 20% of total circulating blood volume⁵⁻⁸. Excessive blood loss not only slows postoperative recovery but also adversely affects surgical outcomes⁹. This is particularly problematic for elderly patients with poor physiological compensation and reserve capacity, as it often leads to severe complications. Additionally, the need for transfusion due to blood loss introduces risks and exacerbates the shortage of blood resources¹⁰⁻¹². Therefore, implementing effective hemostatic measures is crucial to enhancing recovery after TKA^{13,14}.

At present, tranexamic acid (TXA) is the most commonly used hemostatic agent in the TKA perioperative period. TXA that is locally injected prior to the closure of the joint cavity inhibits fibrinolysis and effectively reduces perioperative blood loss by competitively binding to lysine sites on plasminogen¹⁵. Moreover, studies have also shown that the application of TXA does not increase the incidence of thromboembolic complications¹⁶⁻¹⁸. However, the efficacy of TXA application remains suboptimal due to rapid TXA loss and the quick absorption of residual TXA by local vascular cells, resulting in failure to maintain effective drug concentrations, as well as compromised hemostatic efficacy¹⁹⁻²¹. Given that enhanced recovery after surgery protocols demand greater perioperative hemostasis, relying solely on TXA is inadequate, and further optimization and enhancement are necessary.

Absorbable hemostat (AH) is another widely used hemostatic strategy in clinical practice. Unlike the antifibrinolytic mechanism of TXA, AH exerts local hemostatic effects through physical and chemical pathways²²⁻²⁴. Upon contact with blood, AH causes erythrocyte destruction, exposing hemoglobin. The carboxyl groups in AH further oxidize hemoglobin to hematin, releasing ferric ions that accelerate platelet aggregation and adhesion, leading to the formation of a hemostatic thrombus^{22,25}. Additionally, by providing a scaffold for platelet adhesion and aggregation and enabling the formation of a gelatinous mass through its water-absorbing property, AH effectively controls bleeding from capillaries, venules, and arterioles—which can prove challenging when using conventional hemostatic measures²⁶. Given the different hemostatic stages and mechanisms of TXA and AH, their combined use may offer complementary effects. AH can also serve as a scaffold to adsorb TXA, preventing rapid local loss. Unfortunately, there is a lack of solid evidence regarding the combined application of TXA and AH in TKA.

Therefore, we explored the hemostatic effect and safety of TXA combined with AH through in vitro dynamic coagulation tests and a rabbit liver local-bleeding model. Subsequently, we conducted a randomized controlled trial to evaluate the clinical efficacy and safety of TXA combined with AH in reducing perioperative blood loss in TKA. Through this study, we hope to provide new strategies for accelerating patient recovery.

Materials and Methods

The study protocols were approved by the ethics committee of our institution and registered in the Chinese Clinical Trial Registry (ChiCTR2300071664). We later updated the registry information and implemented partial modifications to the outcome classifications. The primary outcomes consisted of direct measures of the efficacy and safety of TXA and AH combination therapy—specifically, perioperative blood loss volume, transfusion requirements, the rate of anemia, hemoglobin and hematocrit levels, and the occurrences of wound complications, thromboembolic events, and other serious adverse events. Indirect measures were classified as secondary outcomes and included functional recovery parameters (joint range of motion, stride speed, Hospital for Special Surgery [HSS] knee score, and visual analogue scale [VAS] pain score), systemic inflammatory and coagulation biomarkers, and other data related to complications. The relevant updated information can be accessed on the registry platform. Additionally, this study was conducted and reported in strict adherence to the Consolidated Standards Of Reporting Trials (CONSORT) statement (see Appendix).

In Vitro Coagulation Test

The procoagulant properties of plain gauze (PG) (Henan Kelong Medical Apparatus & Instruments), PG+TXA (China Resources Double-Crane Pharmaceutical), oxidized cellulose-based AH (SURGICEL; Johnson & Johnson), and TXA+AH were first evaluated by measuring the blood clotting index (BCI). Subsequently, the cellular adhesion of various hemostatic materials was evaluated using scanning electron microscopy (SEM) and lactate dehydrogenase (LDH) enzyme activity assays. Detailed methods are provided in the Appendix.

In Vivo Liver Hemostasis Assay and Biocompatibility Assessment

New Zealand rabbits were immobilized in the supine position following anesthesia. After exposing the liver, a 1-cm-long and 5-mm-deep incision was made on its surface. Hemostatic materials were immediately applied over the incision. Bleeding time and blood loss were meticulously recorded. Two days postoperatively, blood samples were collected for routine blood and biochemical analysis to evaluate the in vivo biocompatibility. See the Appendix for detailed methods.

Patient Recruitment

Patients admitted to our hospital from March to September 2023 were included if they were 18 to 80 years of age; had an American Society of Anesthesiologists (ASA) functional status of I, II, or III; underwent TKA for primary end-stage knee osteoarthritis; and consented to participate in the trial. Exclusion criteria included diagnoses other than primary osteoarthritis, preoperative anemia, an allergy to AH or TXA, anticoagulant use within 7 days before surgery, a history of hematological disorders or venous thrombosis, or a body mass index (BMI) of >35 kg/m². All patients provided informed consent for participation.

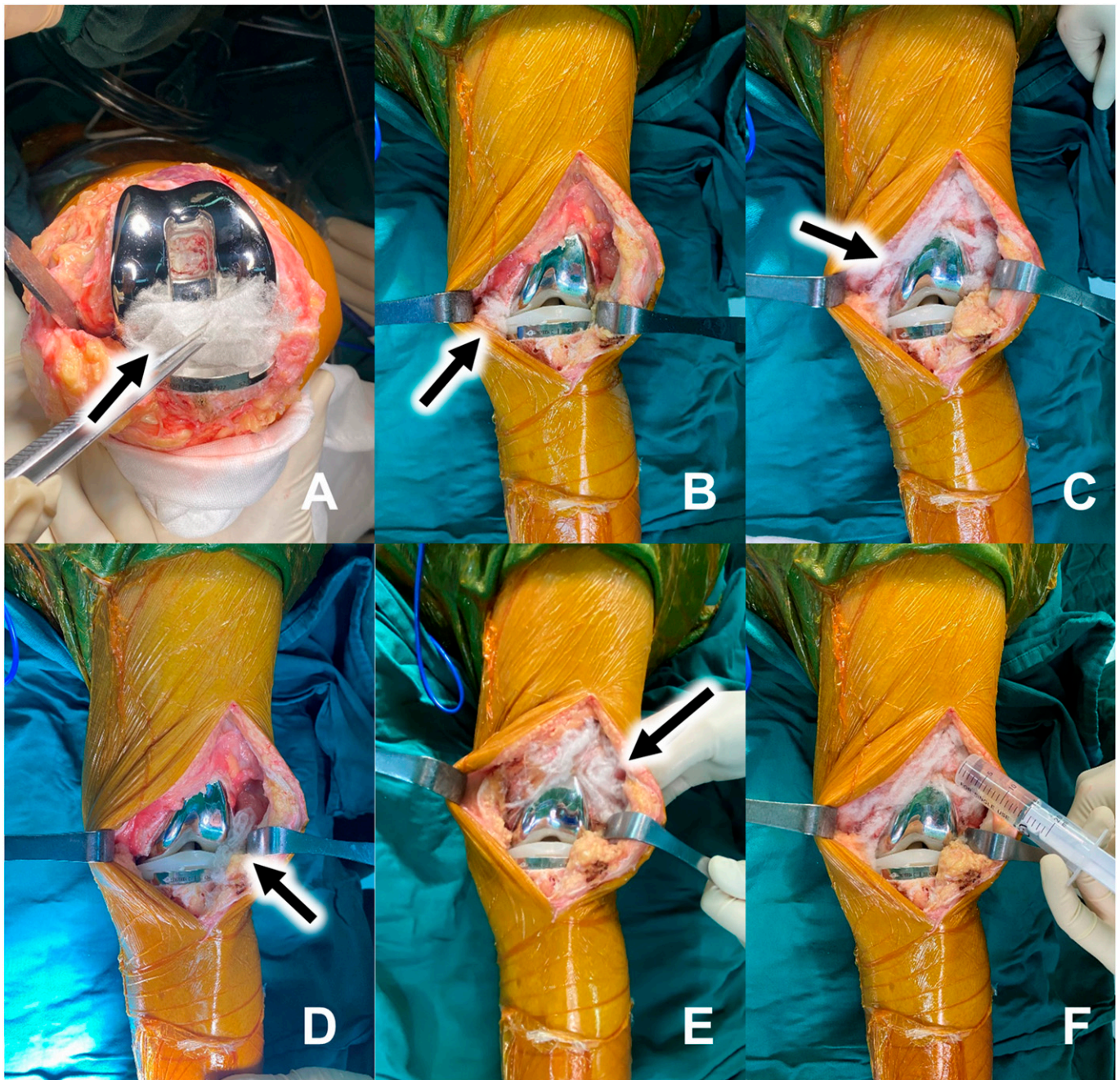


Fig. 1
Images showing the application of AH and TXA during TKA. After implanting the femoral and tibial prostheses, AH was placed on the posterior side of the joint capsule (**Fig. 1-A**), inferiorly (**Fig. 1-B**), on the medial-superior aspect (**Fig. 1-C**), outside the joint capsule inferiorly (**Fig. 1-D**), and on the lateral-superior aspect (**Fig. 1-E**) to achieve 360-degree hemostasis at the surgical site. Subsequently, TXA was injected onto the surface of the AH (**Fig. 1-F**).

Randomization and Blinding

Researcher 1 prepared opaque envelopes containing cards for different groups. On the day of surgery, Researcher 2 randomly drew an envelope and assigned the patient to either the TXA group, the AH group, or the TXA+AH group. Researcher 3 collected the data, and Researcher 4 performed the analysis. Researchers 1, 3, and 4 were blinded to the randomization outcomes and study design.

Surgery and Interventions

All surgeries were performed by the same surgeon using a standard anterior midline incision and a medial parapatellar approach (see Appendix for details)²⁷. A tourniquet was applied for 10 minutes at the time of skin incision and during cementation. After prosthesis implantation, different hemostatic measures were taken according to the group assignment. In the TXA group, 1 g of TXA was applied locally within the joint cavity. In the AH

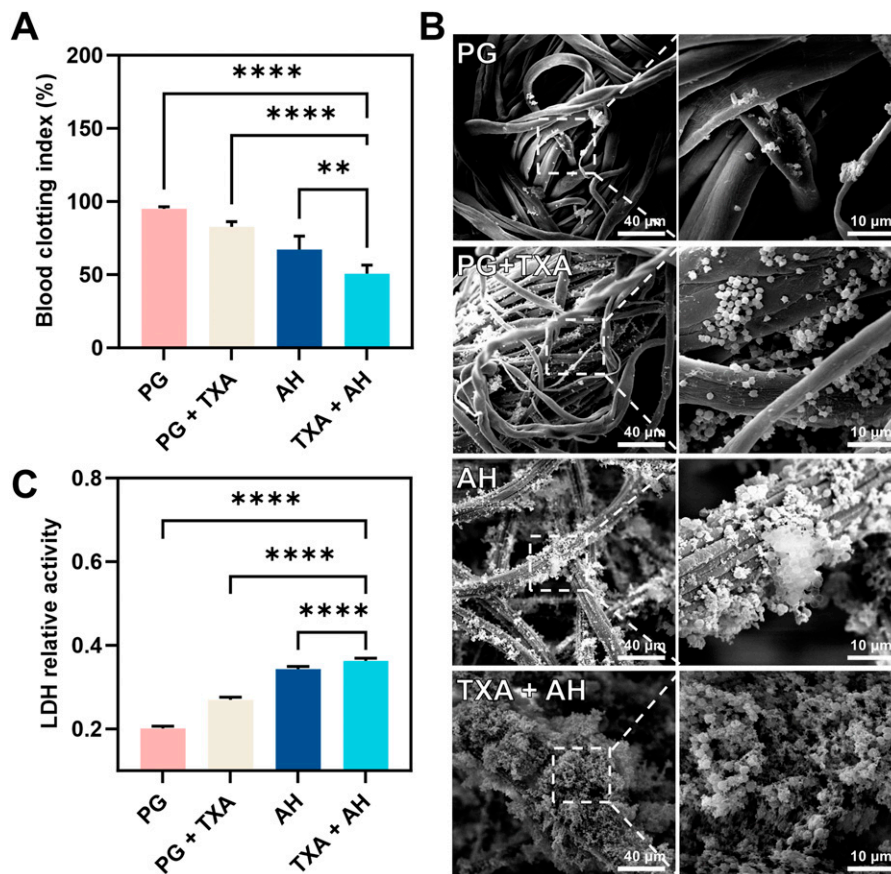


Fig. 2

In vitro hemostatic evaluation of different hemostatic materials. ** $P < 0.01$. **** $P < 0.001$. **Fig. 2-A** BCIs. **Fig. 2-B** SEM observation of coagulation status on the material surfaces. **Fig. 2-C** Determination of LDH activity in cells adhering to the surfaces.

group, a total of 5.1×10.2 cm of AH sheet was positioned posterior, inferior, medial, and superior to the joint capsule (Fig. 1). In the TXA+AH group, AH combined with TXA was used in the knee joint. TXA, at a dose of 0.5 g, was administered intravenously to all three groups at intervals of 3, 6, 9, 12, and 24 hours postoperatively. No postoperative drains were placed in any of the patients. Details regarding perioperative blood and anti-coagulation management are provided in the Appendix.

Outcomes and Follow-up

Demographic data and outcomes were collected from electronic medical records. The primary outcomes included total perioperative, intraoperative, and hidden blood loss; hemoglobin and hematocrit levels; maximum hemoglobin change; the rate of anemia; the number and volume of transfusions; venous thromboembolism; wound-related complications; and other serious adverse events¹⁴. The secondary outcomes included platelet count, activated partial thromboplastin time (APTT), prothrombin time (PT), erythrocyte sedimentation rate (ESR), C-reactive protein (CRP) level, interleukin-6 (IL-6) level, D-dimer level, fibrinogen level, knee circumference, knee mobility, HSS knee score, VAS pain score, stride speed, liver and kidney function, and other adverse events. All patients were followed

for >6 months postoperatively. The measurement methods are detailed in the Supplementary Materials and illustrated in Figure S1 (see Appendix).

Statistical Analysis

Continuous data are presented as the mean $\pm$ standard deviation and were compared using an analysis of variance, with Tukey honestly significant difference (HSD) tests for pairwise comparisons. Categorical data are presented as counts and percentages, and differences among groups were assessed using chi-square or Kruskal-Wallis tests, with the Bonferroni correction for multiple comparisons. A power analysis of the outcomes was also performed (see Appendix Table S1). With a significance level of 0.05 and a power of 0.8, 36 patients were required in each group. Anticipating a loss to follow-up rate of 10%, 40 patients were included in the study. All analyses were conducted using R software (version 4.4.2; R Foundation for Statistical Computing).

Results

Preclinical Laboratory Tests

Laboratory tests were first conducted to evaluate the efficacy and safety of the combination of TXA and AH. The in vitro coagulation test revealed that the BCI of the TXA+AH group

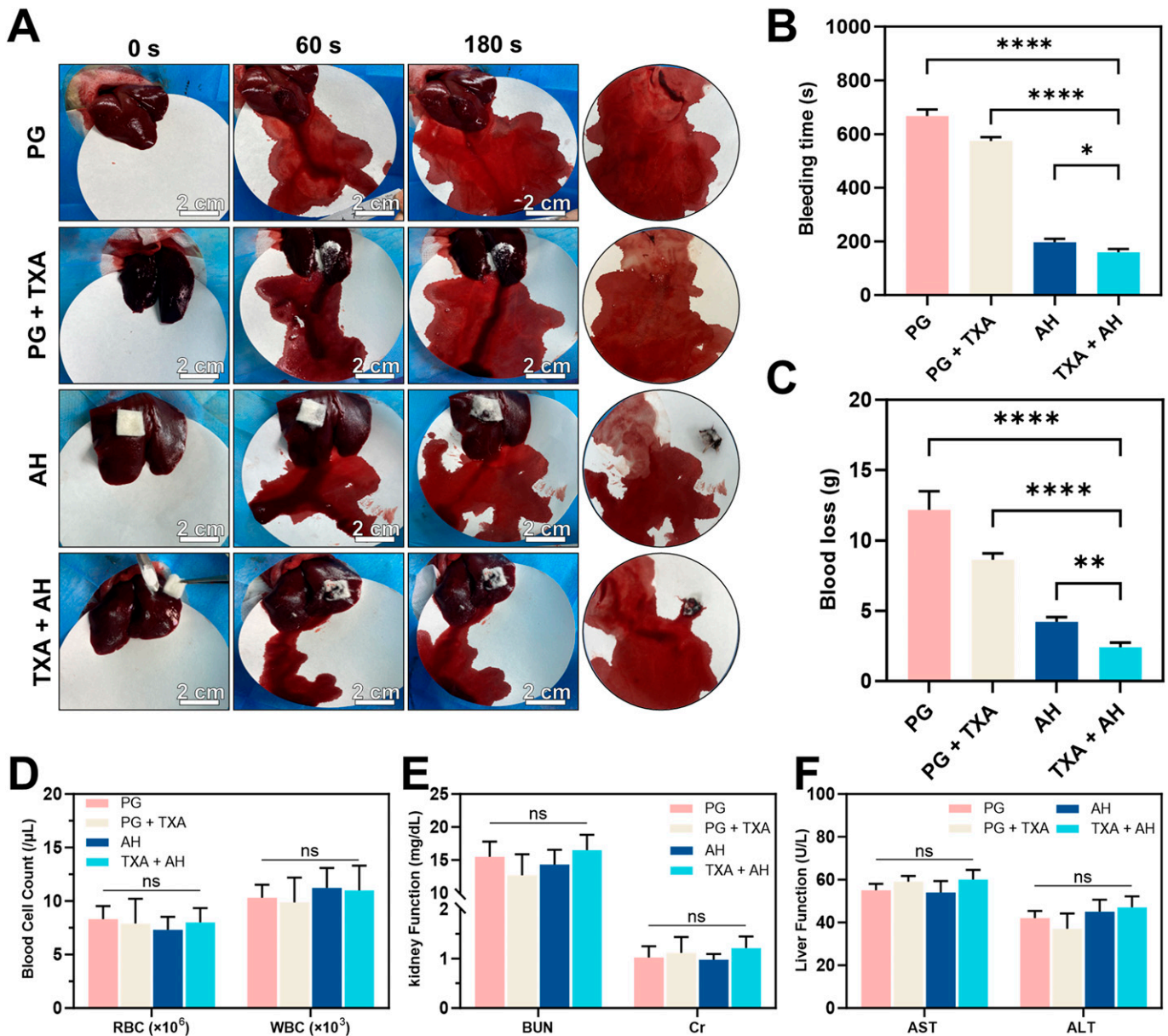


Fig. 3

In vivo liver hemostatic experiments and biocompatibility assessment in rabbits. Photographs of the hemostatic process (**Fig. 3-A**) and bar graphs of the bleeding time (**Fig. 3-B**), blood loss (**Fig. 3-C**), blood-cell count (**Fig. 3-D**), kidney function (**Fig. 3-E**), and liver function (**Fig. 3-F**) at 2 days postoperatively following the use of different hemostatic materials. * $P < 0.05$. ** $P < 0.01$. **** $P < 0.001$. Ns = not significant, RBC = red blood-cell count, WBC = white blood-cell count, BUN = blood urea nitrogen, Cr = creatinine, AST = aspartate aminotransferase, ALT = alanine aminotransferase.

was significantly lower than that of both the AH and PG+TXA groups (**Fig. 2-A**). Additionally, SEM observations of blood coagulation indicated that the TXA+AH group had the highest aggregation of cells and fibronectin content compared with the other groups (**Fig. 2-B**). The intracellular LDH activity measurement also confirmed that the combination of AH and TXA resulted in the greatest cell attachment among the 4 groups (**Fig. 2-C**). These results confirmed that TXA and AH combination therapy exhibited the best coagulative ability in vitro.

Subsequently, we examined the hemostatic efficacy of each group using an in vivo liver bleeding model in rabbits. As shown in **Figure 3-A**, the TXA+AH group demonstrated the best hemostatic results among the groups. Specifically, quantitative analysis revealed that bleeding time and blood loss were significantly reduced in the TXA+AH group compared with the PG, TXA, and AH groups (**Figs. 3-B and 3-C**). Two days postoperatively, the blood test results showed no significant differences in blood-cell counts or kidney and liver function

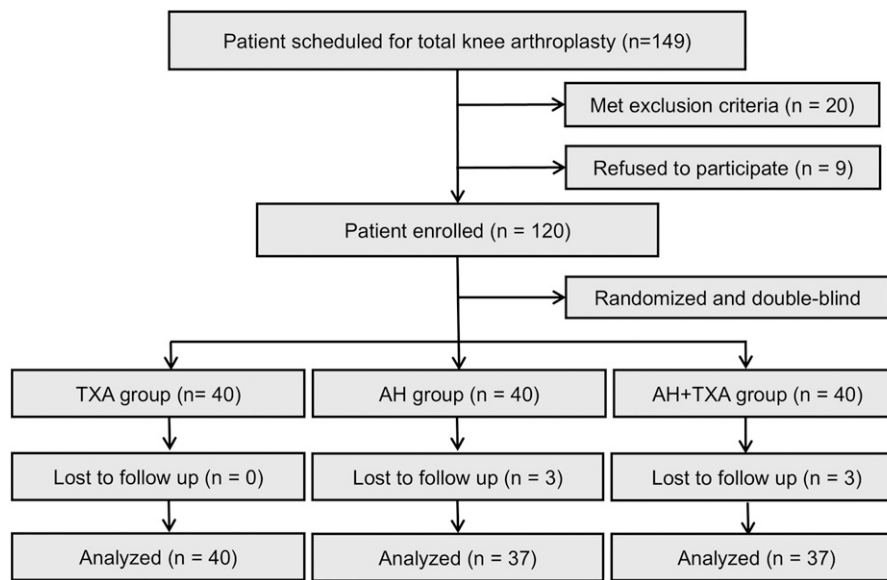


Fig. 4
Flow diagram of patient selection.

across the groups, suggesting good biocompatibility of the combination of TXA and AH (Figs. 3-D, 3-E, and 3-F).

Patient Selection and Baseline Characteristics

Of the 149 patients enrolled, 20 did not meet the inclusion criteria and 9 did not provide consent. The remaining 120 patients were randomized into the TXA, AH, and TXA+AH groups. Two patients in the AH group and 1 in the TXA+AH group refused follow-up due to transportation issues, while 1 and 2 patients in each group, respectively, were lost to follow-up for unclear reasons, resulting in 114 Han Chinese (East Asian) patients eligible for the analysis (Fig. 4). All patients were followed for >6 months. There were no significant differences in baseline characteristics among the groups (Table I).

Clinical Outcomes

Among the primary outcomes, total perioperative and hidden blood loss were significantly lower in the TXA+AH group compared with the TXA and AH groups, whereas intraoperative blood loss was similar between the TXA+AH and AH groups but was notably lower in those groups compared with the TXA group (Figs. 5-A, 5-B, and 5-C). Accordingly, hemoglobin and hematocrit levels on postoperative days 1 and 3 were significantly higher in the TXA+AH group compared with the other groups (Figs. 5-D and 5-E). There were no significant differences in postoperative transfusion rates or volumes. However, patients in the TXA+AH group had a significantly lower incidence of postoperative anemia than the other 2 groups, and the proportion of patients with moderate to severe anemia was also significantly lower in the TXA+AH group. No differences in venous thromboembolism, wound-related complications, or other serious adverse events were observed among the groups (Table II).

With respect to the secondary outcomes, platelet counts were significantly higher in the TXA+AH group than in the TXA group on both postoperative days 1 and 3 and were significantly higher in the AH group than in the TXA group on postoperative day 1 (Fig. 6-A). Nevertheless, the other coagulation-related parameters, including APTT, PT, D-dimer, and fibrinogen, remained consistent among all 3 groups. No differences were observed in liver and kidney function or other adverse events among the groups (see Appendix Fig. S2). In terms of functional recovery, the TXA+AH group demonstrated a significantly higher range of motion than the TXA group (Fig. 6-B). More importantly, the TXA+AH group exhibited a significantly higher HSS score than both the TXA and AH groups on postoperative day 3 and maintained a significant advantage over the TXA group on postoperative day 21 (Fig. 6-C). Similarly, stride speed in the TXA+AH group was significantly higher than that in both the TXA and AH groups on postoperative day 21 and remained significantly superior to that in the TXA group on postoperative day 90 (Fig. 6-D). Finally, the combination of TXA and AH more effectively alleviated knee swelling after surgery compared with TXA alone, while no significant difference in VAS pain was observed among the 3 groups (Table III).

Discussion

During TKA, the implementation of effective and safe hemostatic strategies is crucial to enhancing recovery after surgery^{6,28,29}. Since a single hemostatic measure is insufficient to meet clinical needs, this study explored the combined use of TXA and AH. Through laboratory experiments and a randomized controlled trial, the efficacy and safety of this combination were synergistically validated. The results indicate that the combination of TXA and AH further reduced perioperative blood loss in TKA compared with TXA or AH alone, without increasing

TABLE I Baseline Characteristics of the Patients*

	TXA	AH	TXA+AH	P Value
Demographic information				
Age (yr)	66.10 ± 9.09	66.92 ± 7.74	67.97 ± 7.10	0.595
Sex (M/F) (no. of patients)	9/31	3/34	8/29	0.496
Height (cm)	158.28 ± 7.81	154.84 ± 5.43	156.65 ± 7.59	0.106
Weight (kg)	63.59 ± 10.00	61.69 ± 8.38	63.70 ± 9.64	0.584
Surgical site (L/R) (no. of patients)	20/20	19/18	16/21	0.477
KJC-10 (cm)	40.93 ± 3.81	40.99 ± 3.67	41.91 ± 4.26	0.283
VAS pain	4.82 ± 1.63	4.91 ± 1.29	4.75 ± 1.12	0.735
Preop. laboratory value				
Hb (g/dL)	135.07 ± 11.89	133.4 ± 12.66	136.35 ± 11.18	0.377
HCT (L/L)	0.42 ± 0.03	0.41 ± 0.04	0.42 ± 0.02	0.462
PLT (10 ⁹ /L)	215.68 ± 54.84	218.2 ± 51.04	210.14 ± 65.19	0.325
ESR (mm/h)	16.70 ± 11.38	21.35 ± 11.96	17.16 ± 8.93	0.338
CRP (mg/L)	3.43 ± 1.13	3.17 ± 1.38	3.94 ± 1.62	0.544
IL-6 (pg/mL)	3.00 ± 1.58	3.04 ± 2.02	3.06 ± 1.20	0.908
APTT (s)	25.46 ± 1.90	25.53 ± 1.66	25.61 ± 1.83	0.772
PT (s)	10.86 ± 0.64	10.70 ± 0.48	10.75 ± 0.60	0.502
D-dimer (mg/L FEU)	0.75 ± 0.59	0.76 ± 0.64	0.64 ± 0.55	0.463
Fibrinogen (g/L)	2.96 ± 0.66	3.13 ± 0.62	2.93 ± 0.66	0.403
Comorbidities (no. of patients)				
Hypertension	21 (52.5%)	20 (54.1%)	17 (45.9%)	0.759
Diabetes	10 (25.0%)	9 (24.3%)	11 (29.7%)	0.846
Heart disease	6 (15.0%)	3 (8.1%)	4 (10.8%)	0.630
Other disease	6 (15.0%)	7 (18.9%)	7 (18.9%)	0.871
Intraop. baseline				
Surgical time (min)	90.37 ± 10.65	89.08 ± 9.92	88.92 ± 9.92	0.716
Blood pressure (mmHg)	100.23 ± 12.48	99.46 ± 13.48	101.47 ± 9.38	0.609
Length of wound (cm)	14.87 ± 1.43	14.08 ± 1.58	14.79 ± 1.26	0.886
Preop. knee function				
Range of motion† (deg)	102.87 ± 16.48	103.64 ± 9.32	107.70 ± 13.31	0.839
HSS score	54.12 ± 14.38	55.14 ± 8.55	56.27 ± 9.83	0.590
Stride speed (m/s)	0.82 ± 0.07	0.81 ± 0.08	0.82 ± 0.09	0.989

*Values are given as the mean ± standard deviation or as the number (percent). TXA = tranexamic acid, AH = absorbable hemostat, KJC-10 = knee joint circumference 10 cm above the upper edge of the patella, VAS = visual analogue scale, Hb = hemoglobin, HCT = hematocrit, PLT = platelet count, ESR = erythrocyte sedimentation rate, CRP = C-reactive protein, IL-6 = interleukin-6, APTT = activated partial thromboplastin time, PT = prothrombin time, FEU = fibrinogen-equivalent units, HSS = Hospital for Special Surgery. †Range of motion in knee flexion and extension.

complications. To provide a multidimensional evaluation, and without compromising the conclusions of this study, we implemented a systematic reclassification of the outcomes that established the primary outcomes as those evaluating direct blood management efficacy and safety and that incorporated other indirect measurements as the secondary outcomes. The aims of such modifications were to provide more structured and comprehensive insights regarding the combined application of TXA and AH.

As mentioned previously, TXA is the most commonly used hemostatic agent in TKA. Reports in the literature have indicated that the local injection or intra-articular infusion of

TXA before wound closure can decrease perioperative blood loss, alleviate local swelling at the surgical site, and expedite patient recovery³⁰⁻³². However, rapid TXA loss following direct local application remains inevitable. Besides, when the dose of TXA is too high, its hemostatic effect plateaus and there is an increased risk of postoperative nausea and vomiting and a higher risk of deep venous thrombosis^{33,34}. Therefore, to further enhance the efficacy of TXA, we investigated whether its combined use with AH could further reduce perioperative blood loss in TKA.

Unlike TXA, which inhibits fibrinolysis, AH primarily promotes platelet aggregation and adhesion, enabling the

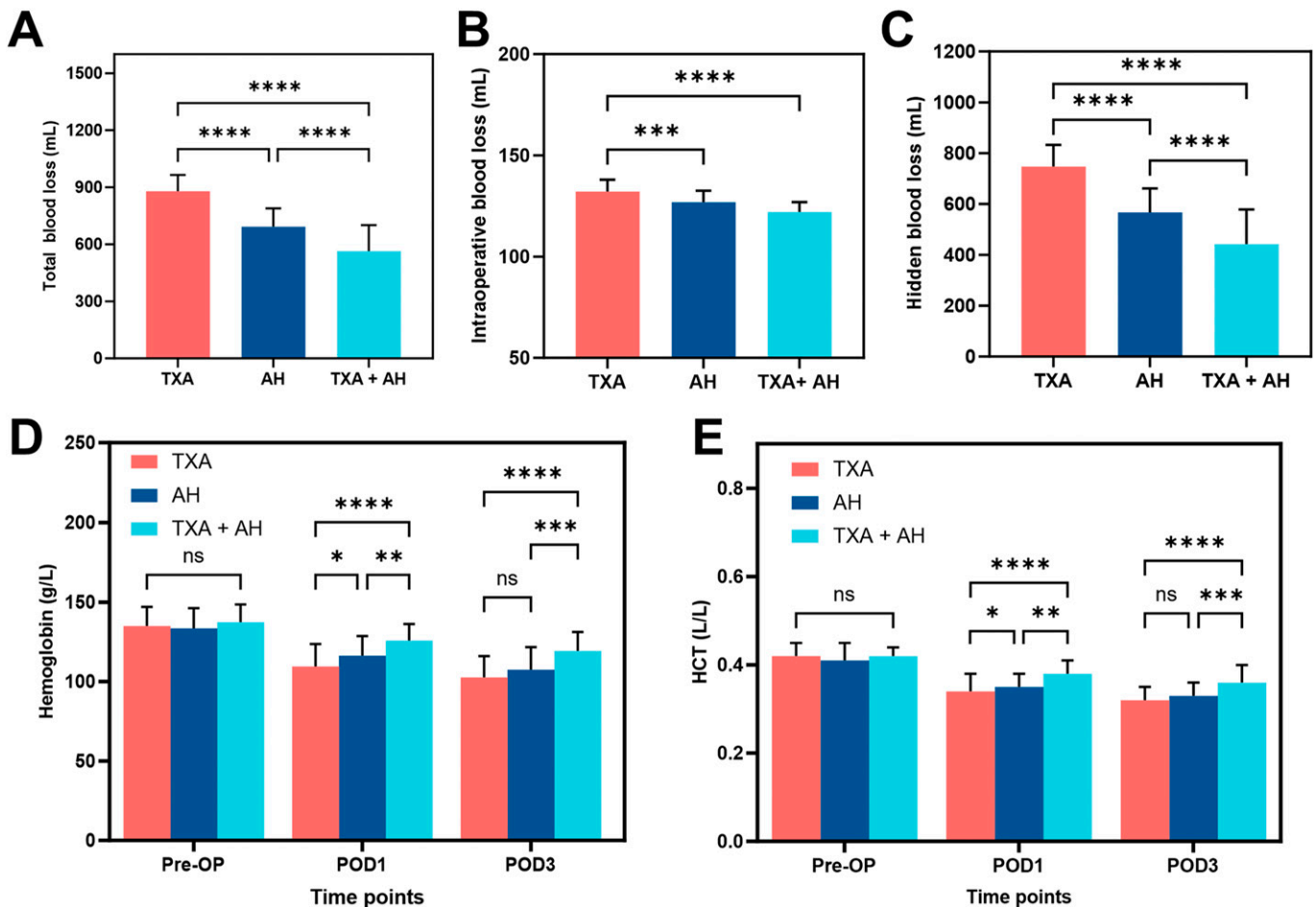


Fig. 5 Evaluation of perioperative blood loss and hemoglobin and hematocrit levels across groups receiving different hemostatic materials. Bar graphs showing total perioperative blood loss (Fig. 5-A), intraoperative blood loss (Fig. 5-B), hidden blood loss (Fig. 5-C), perioperative hemoglobin level (Fig. 5-D), and perioperative hematocrit level (Fig. 5-E). * $P < 0.05$. ** $P < 0.01$. *** $P < 0.005$. **** $P < 0.001$. Ns = not significant, POD = postoperative day.

formation of a hemostatic thrombus. Additionally, AH can induce the release of intracellular contents from ruptured blood cells, further activating the coagulation system²². In a study by Li et al., AH was associated with significantly less perioperative and hidden blood loss compared with a control, with no AH-related complications observed during follow-up²⁴. Currently, oxidized cellulose, hydroxyethyl cellulose, and carboxymethyl cellulose are the main raw materials for AH. Among them, oxidized cellulose has demonstrated the best hemostatic effect, possibly due to its surface charge and the weak acidic environment that it creates, which is more conducive to cellular adhesion and coagulation system activation³⁵. Therefore, oxidized cellulose-based AH was selected for the present study.

We first validated the efficacy of TXA combined with AH in vitro. The BCI in the TXA+AH group was significantly lower than that in the other groups, suggesting that only a minimal number of cells did not participate in coagulation. SEM observations further revealed that the surfaces in the PG and PG+TXA groups had only a small number of cells and

fibers, substantially fewer than the AH group, highlighting the distinct hemostatic mechanisms of TXA and AH. The combination of TXA and AH thus exhibited a complementary effect, promoting early cellular aggregation and adhesion while inhibiting fibrinolysis in the later stages, resulting in a remarkable increase in cell and fiber components on the material surface. Using intracellular LDH activity measurements, we quantitatively confirmed that the TXA+AH group had significantly enhanced cellular adhesion during hemostasis, outperforming the other groups. We further validated the hemostatic effect of TXA combined with AH in vivo using a liver bleeding model that is commonly employed for coagulation function tests³⁶. The results similarly confirmed that the TXA+AH group had a significantly shorter bleeding time and less blood loss than the other groups.

To further investigate the clinical efficacy of TXA combined with AH in reducing perioperative blood loss in TKA, we conducted a randomized controlled trial. The results showed that the combination was associated with significantly reduced perioperative total and hidden blood loss compared with the

TABLE II Perioperative Blood Loss and Complications*

Variable	TXA	AH	TXA+AH	P Value
Blood loss				
Total (mL)	879.35 ± 85.62	692.70 ± 96.06†	564.51 ± 136.26† ‡	<0.001
Intraop. (mL)	132.13 ± 5.84	126.95 ± 5.66†	125.97 ± 4.99†	0.002
Hidden (mL)	747.23 ± 85.32	566.62 ± 94.86†	442.54 ± 136.22† ‡	<0.001
1st postop. day				
Hb (g/L)	109.47 ± 14.16	116.05 ± 12.35†	125.86 ± 10.36† ‡	<0.001
HCT (L/L)	0.34 ± 0.04	0.35 ± 0.03†	0.38 ± 0.03† ‡	<0.001
3rd postop. day				
Hb (g/L)	102.67 ± 13.28	107.54 ± 14.09†	119.35 ± 11.87† ‡	<0.001
HCT (L/L)	0.32 ± 0.03	0.33 ± 0.03	0.36 ± 0.04† ‡	<0.001
Maximum Hb change (g/L)	32.93 ± 6.05	25.16 ± 6.11†	18.11 ± 7.37† ‡	<0.001
IV fluid amount				
Day of operation (mL)	2,889.75 ± 99.45	2,897.84 ± 97.33	2,895.41 ± 89.21	0.930
3rd postop. day (mL)	100.00 ± 11.09	98.92 ± 7.74	97.77 ± 7.60	0.565
Anemia				<0.001§
Normal	10 (25.0%)	16 (43.2%)	26 (70.3%)	
Mild (Hb ≥90 and <130/120 g/L)#	23 (57.5%)	16 (43.2%)	11 (29.7%)	
Moderate (Hb ≥60 and <90 g/L)	6 (15.0%)	5 (13.5%)	0 (0%)	
Severe (Hb <60 g/L)	1 (2.5%)	0 (0%)	0 (0%)	
Transfusions				
Number	1 (2.5%)	0	0	0.168
Volume (units)	1.5	0	0	0.173
Venous thromboembolism (no. of patients)				
Deep venous thrombosis	2 (5.0%)	1 (2.7%)	1 (2.7%)	0.816
Pulmonary embolism	0	0	0	–
Wound complication (no. of patients)				
Seepage	4 (10%)	3 (8.1%)	2 (5.4%)	0.755
Infection	0	0	0	–
Non-healing	0	0	0	–
Other serious adverse event (no. of patients)				
Shock	0	0	0	–
Death	0	0	0	–

*Values are given as the mean ± standard deviation or as the number (percent). TXA = tranexamic acid, AH = absorbable hemostat, Hb = hemoglobin, HCT = hematocrit, IV = intravenous. †Significantly different from the TXA group. ‡Significantly different from the AH group. §The AH + TXA group significantly differed from both the TXA and AH groups. #90~130 g/mL for male patients and 90~120 g/mL for female patients.

other 2 groups and that it also reduced intraoperative blood loss compared with the TXA group. Additionally, higher hemoglobin and hematocrit levels were observed in the TXA+AH group compared with both the TXA and AH groups on postoperative days 1 and 3. These findings indicate that the combination of TXA and AH covers the entire process, from initial hemostasis to the inhibition of rapid postoperative fibrinolysis, and demonstrates superior hemostatic efficacy. Moreover, the TXA+AH group exhibited superior reductions in knee joint swelling and improvements in range of motion relative to the TXA group and achieved higher HSS scores on postoperative day 3 and higher stride speed on postoperative day 21 compared with both the TXA

and AH groups, collectively indicating that the combination accelerates early postoperative functional recovery. We believe that there are a few potential reasons for these findings. First, the combination of TXA and AH effectively reduced perioperative blood loss, preserving the tissue oxygen supply and consequently accelerating tissue repair. Second, a reduction in hidden blood loss has been shown to alleviate joint swelling and pain, improving joint mobility and functional outcomes⁴⁰. Finally, prior research has demonstrated that the amount of blood loss is negatively correlated with the postoperative mental state of the patient, which can lead to better and more active participation in postoperative rehabilitation⁴¹.

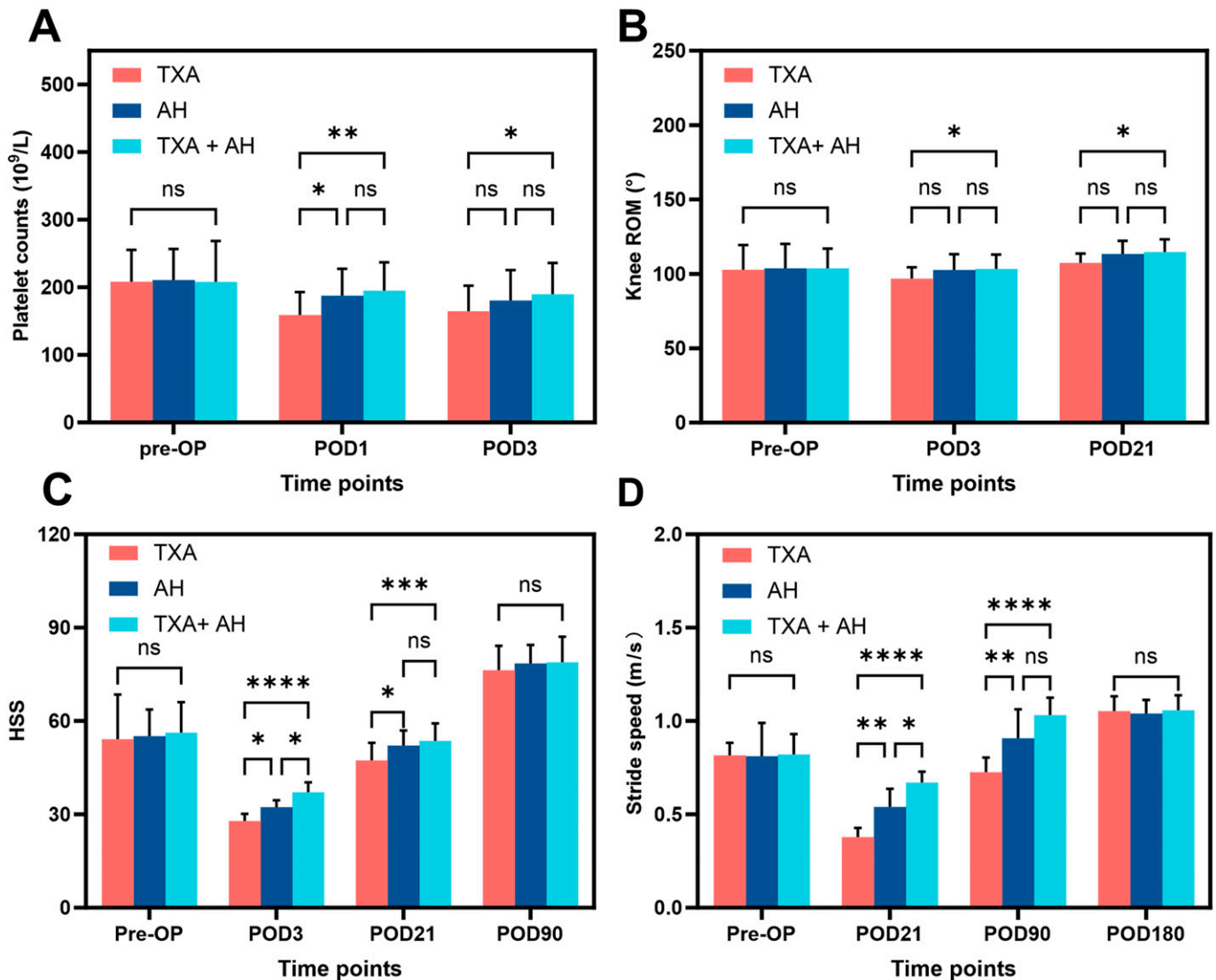


Fig. 6 Evaluation of platelet count and postoperative functional recovery. Bar graphs showing platelet count (Fig. 6-A), knee mobility (Fig. 6-B), HSS scores (Fig. 6-C), and stride speed (Fig. 6-D) across groups receiving different hemostatic materials. * $P < 0.05$. ** $P < 0.01$. *** $P < 0.005$. **** $P < 0.001$. Ns = not significant, POD = postoperative day.

Regarding safety, we found that the combination did not increase the risks of deep venous thrombosis and pulmonary embolism, which are the most serious complications of TXA use. Additionally, since AH serves as a scaffolding material to bind TXA through physical adsorption and intermolecular forces such as ionic and hydrogen bonds, the combination could avoid the rapid loss of TXA during surgery, thereby preventing the need for high-dose TXA, as well as the related side effects. Nevertheless, combining TXA with AH may alter the pharmacokinetic characteristics of TXA itself. Herein, we verified that such a combination caused no additional adverse effects on key organs (the liver and kidney) involved in drug metabolism. Previous studies have linked AH-related complications to factors such as AH size, local blood flow, placement location, and whether the AH is removed after hemostasis³⁷⁻³⁹.

In a review of the application of AH in surgery, Franceschini reported that surgical site infections, allergic reactions, foreign body reactions, and nerve compression are common postoperative issues³⁹. In the present study, no significant abnormalities were observed in inflammatory markers or wound healing in the AH and TXA+AH groups, indicating that local, rational use of AH, with strict adherence to usage indications, is safe when combined with TXA.

This study has limitations. First, it only included patients with primary knee osteoarthritis and did not evaluate other types of patients undergoing TKA. Second, as a single-center study, the sample size was relatively small, and thus larger sample sizes and multicenter studies will be necessary in the future. Third, due to ethical considerations, the clinical trial grouping in this study involved adding AH to the hemostatic

TABLE III Inflammation, Coagulation Function, and Knee-Related Parameters After Surgery*

Variable	TXA	AH	TXA+AH	P Value
1st postop. day				
Indicator of inflammation				
ESR (mm/h)	25.20 ± 14.00	23.62 ± 13.75	22.43 ± 15.25	0.344
CRP (mg/L)	41.31 ± 25.60	52.14 ± 30.07	44.11 ± 22.38	0.142
IL-6 (pg/mL)	164.45 ± 120.36	149.28 ± 80.93	142.34 ± 98.58	0.543
Coagulation function				
APTT (s)	26.52 ± 2.15	26.17 ± 1.84	26.42 ± 2.05	0.746
PT (s)	11.85 ± 0.65	11.59 ± 0.65	11.63 ± 0.69	0.360
PLT (10 ⁹ /L)	158.97 ± 33.87	187.81 ± 39.68†	197.29 ± 42.64†	0.005
D-dimer (mg/L FEU)	3.00 ± 2.46	2.62 ± 2.02	2.37 ± 1.61	0.404
Fibrinogen (g/L)	3.12 ± 0.81	3.56 ± 1.94	3.14 ± 0.67	0.385
KJC-10 (cm)	46.88 ± 4.35	45.09 ± 4.11†	43.85 ± 4.01†	0.024
3rd postop. day				
Indicator of inflammation				
ESR (mm/h)	57.42 ± 22.04	59.78 ± 25.80	54.35 ± 23.65	0.522
CRP (mg/L)	128.61 ± 50.53	133.68 ± 40.27	120.56 ± 52.48	0.451
IL-6 (pg/mL)	58.61 ± 37.05	71.74 ± 61.81	64.04 ± 47.75	0.459
Coagulation function				
APTT (s)	27.97 ± 2.53	27.99 ± 2.89	27.97 ± 1.84	0.838
PT (s)	11.86 ± 0.95	11.64 ± 0.93	11.55 ± 0.61	0.347
PLT (10 ⁹ /L)	164.65 ± 37.67	180.51 ± 44.86	189.92 ± 45.95†	0.036
D-dimer (mg/L FEU)	2.48 ± 1.89	2.34 ± 1.72	2.39 ± 1.42	0.766
Fibrinogen (g/L)	6.36 ± 1.22	6.24 ± 1.09	6.04 ± 1.41	0.496
VAS pain	3.25 ± 1.08	2.97 ± 0.93	2.75 ± 0.83	0.556

*Values are given as the mean ± standard deviation. TXA = tranexamic acid, AH = absorbable hemostat, ESR = erythrocyte sedimentation rate, CRP = C-reactive protein, IL-6 = interleukin-6, APTT = activated partial thromboplastin time, PT = prothrombin time, PLT = platelet count, FEU = fibrinogen-equivalent units, KJC-10 = knee joint circumference 10 cm above the upper edge of the patella, VAS = visual analogue scale. †Significantly different from the TXA group.

regimen already in use at our hospital, with no blank control group, and the hemostatic effect in the TXA+AH group was compared with that in the TXA-only group. Fourth, the present report implemented a reclassification of outcome measures and incorporated additional outcomes in accordance with peer-review recommendations. These modifications aimed to more structurally and comprehensively assess the efficacy and safety of the combination of TXA and AH. However, although some positive outcomes (such as those regarding anemia and blood tests) were reclassified as primary endpoints, these changes were unlikely to introduce additional reporting bias. Finally, no drainage tubes were placed postoperatively, and hidden blood loss has been overestimated, which may have affected the interpretation of the results.

In summary, this study innovatively demonstrated that the combination of TXA and AH reduced perioperative blood loss in TKA and accelerated early postoperative recovery without increasing postoperative complications. This combination may further enhance blood management efficiency in joint surgeries and may be applied to other fields.

Appendix

 Supporting material provided by the authors is posted with the online version of this article as a data supplement at [jbjs.org \(http://links.lww.com/JBJS/1789\)](http://links.lww.com/JBJS/1789). ■

Note: The authors thank Ms. Chen Maojia and Mr. Rui Wang of the Experimental Animal Center, West China Hospital, Sichuan University, for their invaluable assistance with the animal experiments.

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